

PARTICIPANT INFORMATION SHEET

A Study on the Usability of Differential Privacy Libraries

Before you decide whether or not to participate, we would like you to understand why the research is being conducted and what it will involve. Please ask us questions if anything is unclear.

What is the purpose of the project? This research aims to explore the usability of Differential Privacy (DP) libraries to identify what struggles developers are likely to face when using them. To do this we are going to watch and record you while you are solving a task using the DP libraries and narrating your thought processes.

Why have I been invited to participate? You have been invited to participate as someone interested who has experience with Python programming and Data Privacy Techniques.

The eligibility criteria to participate in the study To be able to participate in this study, you must have a good understanding of Python programming and basic understanding of Privacy Techniques. This study is designed for developers who are both experienced and novice in implementing Privacy techniques. During the recruitment process, you will be asked screening questions to determine what kind of programming and Privacy implementation experience do you have to participate in this study.

Do I have to take part? It is up to you to decide whether you wish to participate in the project. We will describe the study and go through this information sheet with you before you participate, and answer any questions you might have. If you agree to take part, we will then ask you to sign a consent form. You are free to withdraw at any time, without giving a reason.

At the end of the session recording will be sent for transcription, and then anonymised by removing individual names, and identifying information. Once the recording is transcribed and anonymised it will not be possible to withdraw. *If you wish to withdraw after the session it is important to notify us as soon as possible (within 7 days).*

What will happen to me if I take part and what will I have to do?

- In this study we will provide you with a dataset and ask you to use 3 differential privacy libraries (namely: DiffPrivLib, PyDP and OpenDP) to count the number of individuals in >50K salary range and Average age for different education levels.
- The session will take between 1.5 - 2 hours. You will be given 1.5 hours to solve the task and an additional 15-30 minutes to go through the study and fill out the related forms.
- We will ask you to narrate your thought processes whilst solving the task.
- The audio of the interview will be recorded using Microsoft Teams/Zoom.
- You will have access to the internet and you can search any resources you may need to fulfil this task.
- We will ask you to share your screen to be able to observe how you solve the task, however, the screen will not be recorded.

- The recorded audio will be transcribed and then anonymised, before being analysed with a standard qualitative method for identifying learning outcomes (thematic analysis). After transcription, the audio recordings will be securely destroyed.

What are the possible disadvantages and risks involved in taking part in the project? There are no risks to taking part in this study.

What are the possible benefits of taking part? You will be reimbursed for your time with a £100 voucher if recruited on completion of the interview.

Will my participation in this project be kept confidential? An audio recording of your session will be transcribed by us. This transcription will then be anonymised by removing names and personally identifiable information. The audio recording will be stored securely in University of Bristol controlled storage (OneDrive). Anonymised transcriptions will be made available as open access data as part of the project. Audio recordings will be securely destroyed once the transcriptions have been verified.

What is open access data? Open access data means that data are made available, free of charge, to anyone interested in the research, or who wishes to conduct their own analysis of the data. We will therefore have no control over how these data are used. However, all data will be anonymised before it is made available and therefore there will be no way to identify you from the research data.

What will happen to the results of the research project? Anonymised transcriptions and analysis will be made available as open access research and may be written up for publication at a later date.

Who has reviewed the study? The study has been reviewed by the primary investigator, and the Faculty of Engineering Research Ethics Committee.

Further information and contact details For further information about this study please contact Ravi Mahankali (ravi.mahankali@bristol.ac.uk).

If participants have any concerns related to their participation in this study please speak to the Faculty of Engineering Research Ethics Committee, via Liam McKervey, Research Governance and Ethics Officer (Tel: 0117 331 7472 email: Liam.McKervey@bristol.ac.uk)